SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date	6 July2026
Team ID	
Project Name	Plugging into the Future :An Exploration of Consumption Patterns
Maximum Marks	3Marks

Project Executable Files

Step1:Submission Checklist

S.No	Item to Submit	Submitted(Yes/No/ NA)
1	Complete source code(all files and folders)	Yes
2	README/Setup Guide(instructions to run the project)	Yes
3	Requirements .txt/pack age. Njs on/dependency file	Yes
4	Data bases chema /seed files(if applicable)	Yes
5	Environment configuration file (. Env . example or similar)	NA
6	Deployed application URL(if hosted)	**No** (Local Deployment)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Docker file /Contain erization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walk through (if required)	Yes

Step2 :File/ Folder Structure

```
Plugging into the Future : An Exploration of Electricity consumption Patterns /
│
├──1.Brainstorming & Ideation /          - Initial project ideas and problem identification.
│ ├──Brainstorming . pdf              - Brainstorming discussion.
│ ├──Literature Survey . pdf           - Existing research review.
│ ├──Problem Statement . pdf            - Defines the project problem.
│ └──Empathy Map.pdf                   User needs and pain points.
│
├──2.Requirement Analysis/               - Project requirements.
│ ├──Customer Journey.pdf                - User interaction flow.
│ ├──Data Flow Diagram . pdf             -System data flow.
│ ├──Functional Requirements . pdf         -System function alities.
│ └──Technology Stack . pdf              - Technologies used.
│
├──3.Project Design Phase /              -System design documents.
│ ├──Solution Architecture . pdf         - Overall system architecture.
│ ├──Problem Solution Fit . pdf          - Problem and solution mapping.
│ ├──Proposed Solution . pdf             -Proposed  system details.
│ └──Sprint Planning.pdf              - Sprints chedule.
│
├──4.Project Planning Phase/              - Project planning documents.
│ ├──Mile stone Planning . pdf           - Project mile stones.
│ ├──Project Planning . pdf            - Development plan.
│ └──Sprint Schedule . pdf             - Sprint time line.
│
├──5.Project Development Phase/            - Source code implementation.
│ └──Source Code/
│     ├──static /                - Static resources.
│     │ └──style . css             - Application styling.
│     │
│     ├──templates /               - HTML web  pages.
│     │ ├──index . html             -Home page.
│     │ ├──result . html            - Prediction result page.
│     │ ├──history . html            - Prediction history page.
│     │ ├──dashboard . html             - Analytics dash board.
│     │ └──login . html             - Login page.
│     │
│     ├──app . py                 -Flask application.
│     ├──main . py                 -Model training script.
│     ├──model . pkl                -Trained ML model.
│     ├──electricity_usage.db         - SQ Lite data base.
│     ├──consumption_data.csv          - Power consumption  dataset.
│     ├──historical_records.csv       -  Historical usage dataset.
│     ├──requirements . txt          - Required Python pack ages.
│     └──README . md                 -Project instructions.
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├──6.Project Testing/            -Testing documents.
│ ├──Test Cases . pdf             -Test case details.
│ ├──Application Building.pdf       -Build process.
│ └──Output Screenshots .pdf          -Application screen shots.
│
├──7.Project Documentation/         -Project documentation.
│ ├──Final Report.pdf              -Final project report.
│ └──Project Presentation.pptx        -Project presentation.
│
├──8.Project Demonstration/          -Demon stration files.
│ ├──Demo Video.mp4                -Project demo video.
│ └──Presentation.pptx              -Demonstration slides.
│
├──README.md                       -Project over view.
├──requirements . txt              -Depend encylist.
└──.giti gnore                   -Gitignore configuration.
```

Step3:Deployment / Access Details

Field	Details
Hosted/Deployed URL	Not Deployed (Runson Local Flask Server)
Login Credentials (Demo)	Not Applicable
Platform/ Hosting Provider	Local Flask Development Server (Python + Flask+ SQ Lite)
Repository Link	https://github.com/vinukondadaniel4-sketch/Vinukonda-varshitha-.git
Demo Video Link	

Step 4:Run Instructions

Pre-requisites : Python 3.10 or above, Visual Studio Code, Flask, NumPy, Pandas, Scikit-learn,Matplotlib,Seaborn,SQLite3,andawebbrowser(GoogleChromeorMicrosoftEdge).
Step1: Open the Electricity_Consumption_Project folder in Visual Studio Code

Step2: Create and activate the virtual environment using the commands:"python-mvenv .venv"and".venv\Scripts\activate"or".\.venv\Scripts\Activate.ps1".

Step3: Install all the required Python libraries using the command:"pip install –r requirements.txt".

Step4: Start the Flask application using the command:"python pp.py".

Step 5: Open your web browser and navigate

to**http://127.0.0.1:5000/**.

Step 6: Enter the region and usage details in the form, click Predict Consumption, view the predicted electricity usage along with analytics, access the Prediction History to view saved records, and open the Dashboard to view consumption statistics and trends.

Step5:Known Issues/Limitations

S.No	Known Issue / Limitation	Work around/Status
1	Application currently runs only on the local Flask Development server.	Deploy the application to a cloud platform such as Render ,Railway, or Azure.
2	SQ Lite database issue itable only for small to medium-sized applications.	Upgrade to My SQL or Postgre SQL for large-scale deployment.
3	The application require as local Python environment to execute.	Package the application using Docker or deploy it as a web service for easier access.
4	Basic login authentication is implemented.	Add password encryption , email verification , androle -basedaccess control.